

Thank you!

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PatientProfiles

Adding patient-level characteristics to OMOP CDM tables

2026-06-25

What you will be able to do

- Add patient-level characteristics to cohort and CDM tables.
- Check whether records are inside observation and quantify lookback/follow-up time.
- Create features from intersections with cohorts, concepts, and OMOP CDM tables.

Today's workflow

1. Connect to a synthetic OMOP CDM database.
2. Create two small cohorts for live examples.
3. Add demographics and observation-period features.
4. Build intersection features.



Settings

PatientProfiles starts with a table that has a person identifier and, usually, an index date.

It returns the same table with extra columns.



Tip

For cohort tables, the default index date is **cohort_start_date**. For CDM clinical tables, set the relevant date column explicitly.

Setup

Load packages and create the CDM object

- Load the required packages

```
library(PatientProfiles)
library(CDMConnector)
library(CohortConstructor)
library(duckdb)
library(dplyr)
library(here)
library(omock)
library(DBI)
```

- Create a CDM object from the **GiBleed** synthetic dataset

```
con <- dbConnect(drv = duckdb(dbdir = here("Databases", "GiBleed.duckdb")))
```

```
cdm <- cdmFromCon(
  con = con,
  cdmSchema = "main",
  writeSchema = "results",
  writePrefix = "pp_"
)
```

Create a cohort

Using **CohortConstructor** we will create:

- **my_cohort**: people with arthritis-related records.
- **medications**: people with heparin records.

```
cdm$my_cohort <- conceptCohort(  
  cdm = cdm,  
  conceptSet = list(arthritis = c(80180, 80809)),  
  name = "my_cohort"  
)  
  
cdm$medications <- conceptCohort(  
  cdm = cdm,  
  conceptSet = list(heparin = 1367571),  
  name = "medications"  
)
```


Add demographics

addSex()

Use `addSex()` when the table contains `person_id` or `subject_id` and you want the person's sex as a new column:

```
cdm$my_cohort |>
  addSex(
    sexName = "sex", # name of the new column (default = "sex")
    missingSexValue = "None" # label for missing gender_concept_id (default = "None")
  )
```

A query: ?? x 5

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	sex
	<int>	<int>	<date>	<date>	<chr>
1	1	6	2005-07-13	2005-07-13	Female
2	1	123	1986-03-24	1986-03-24	Male
3	1	129	2014-03-21	2014-03-21	Male
4	1	16	2013-03-21	2013-03-21	Female
5	1	65	2009-06-03	2009-06-03	Female
6	1	74	2015-08-12	2015-08-12	Female
7	1	42	1950-05-16	1950-05-16	Female
8	1	187	1984-11-29	1984-11-29	Male
9	1	18	2009-03-21	2009-03-21	Female

addAge()

Use `addAge()` to calculate age at an index date. The table must contain `person_id` or `subject_id`:

```
cdm$my_cohort |>
  addAge(
    indexDate = "cohort_start_date", # date to compute age (default = "cohort_start_date")
    ageName = "age", # name of the age column (default = "age")
    ageMissingMonth = 1, # Month for individuals with missing month (default = 1)
    ageMissingDay = 1, # Day for individuals with missing day (default = 1)
    ageImposeMonth = FALSE, # Whether to impose default month to all individuals (default = FALSE)
    ageImposeDay = FALSE, # Whether to impose default day to all individuals (default = FALSE)
    ageUnit = "years" # 'years', 'months' or 'days'.
  )
```

A query: ?? x 5

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	age
	<int>	<int>	<date>	<date>	<int>
1	1	84	2014-03-29	2014-03-29	40
2	1	406	1996-06-27	1996-06-27	44
3	1	1758	2002-09-28	2002-09-28	42
4	1	1820	2014-08-17	2014-08-17	36
5	1	2309	2016-10-10	2016-10-10	43
6	1	4247	2018-03-30	2018-03-30	35
7	1	4643	1944-06-22	1944-06-22	35
8	1	4662	1994-06-10	1994-06-10	43
9	1	648	1971-07-25	1971-07-25	38

addAge()

Add age groups

```
cdm$drug_exposure |>
  addAge(
    indexDate = "drug_exposure_start_date",
    ageGroup = list(c(0, 39), c(40, Inf))
  ) |>
  select("person_id", "drug_exposure_start_date", "age", "age_group")
```

A query: ?? x 4

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	person_id	drug_exposure_start_date	age	age_group
	<int>	<date>	<int>	<chr>
1	573	1960-04-09	0	0 to 39
2	1332	2010-10-06	67	40 or above
3	576	2017-10-25	51	40 or above
4	4550	2000-11-13	38	0 to 39
5	3895	1982-03-30	58	40 or above
6	3199	1975-08-19	2	0 to 39
7	476	2008-05-31	40	40 or above
8	1154	1971-05-16	0	0 to 39
9	186	1976-05-31	4	0 to 39
...	...	...	...	...

addAge()

You can create several labelled age-group columns in one call:

```
cdm$drug_exposure |>
  addAge(
    indexDate = "drug_exposure_start_date",
    ageGroup = list(
      "age_group" = list("<40" = c(0, 39), ">=40" = c(40, Inf)),
      "category" = list("kids" = c(0, 17), "adults" = c(18, Inf))
    )
  ) |>
  select("person_id", "drug_exposure_start_date", "age", "age_group", "category")
```

A query: ?? x 5

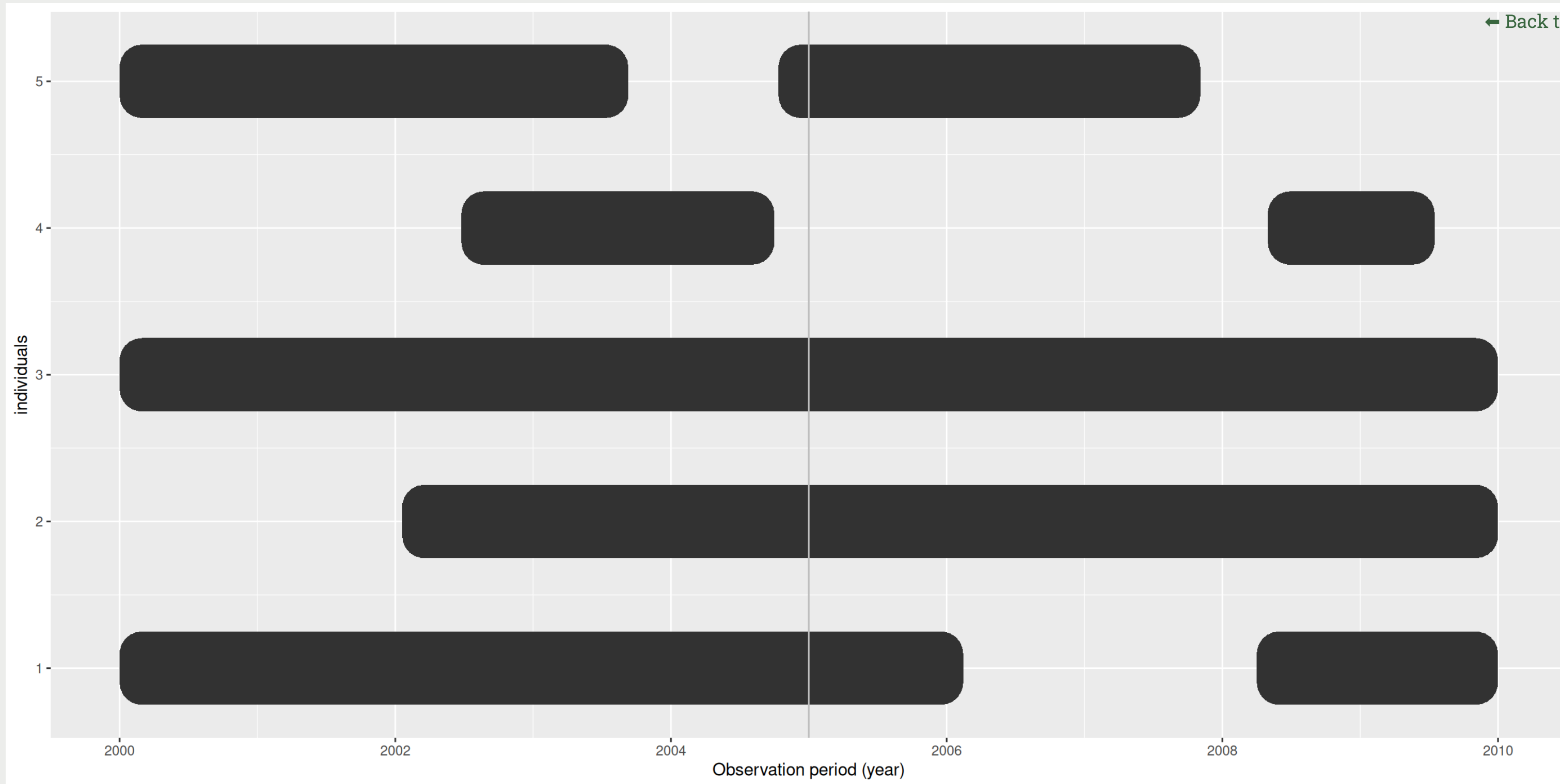
Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

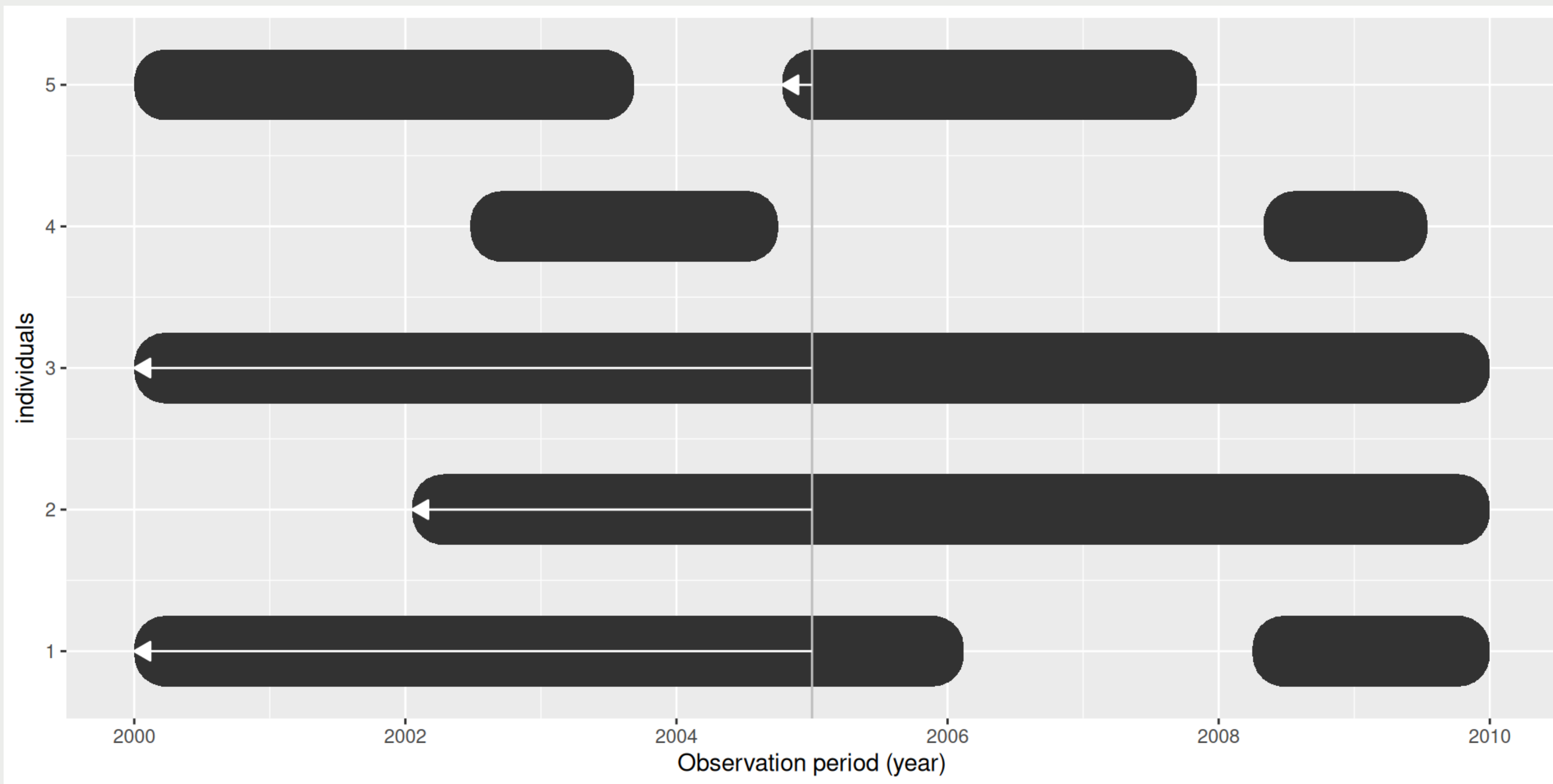
	person_id	drug_exposure_start_date	age	age_group	category
	<int>	<date>	<int>	<chr>	<chr>
1	573	1960-04-09	0	<40	kids
2	1332	2010-10-06	67	>=40	adults
3	576	2017-10-25	51	>=40	adults
4	4550	2000-11-13	38	<40	adults
5	3895	1982-03-30	58	>=40	adults
6	3199	1975-08-19	2	<40	kids
7	476	2008-05-31	40	>=40	adults
8	1154	1971-05-16	0	<40	kids
9	186	1976-05-31	4	<40	kids

addPriorObservation()

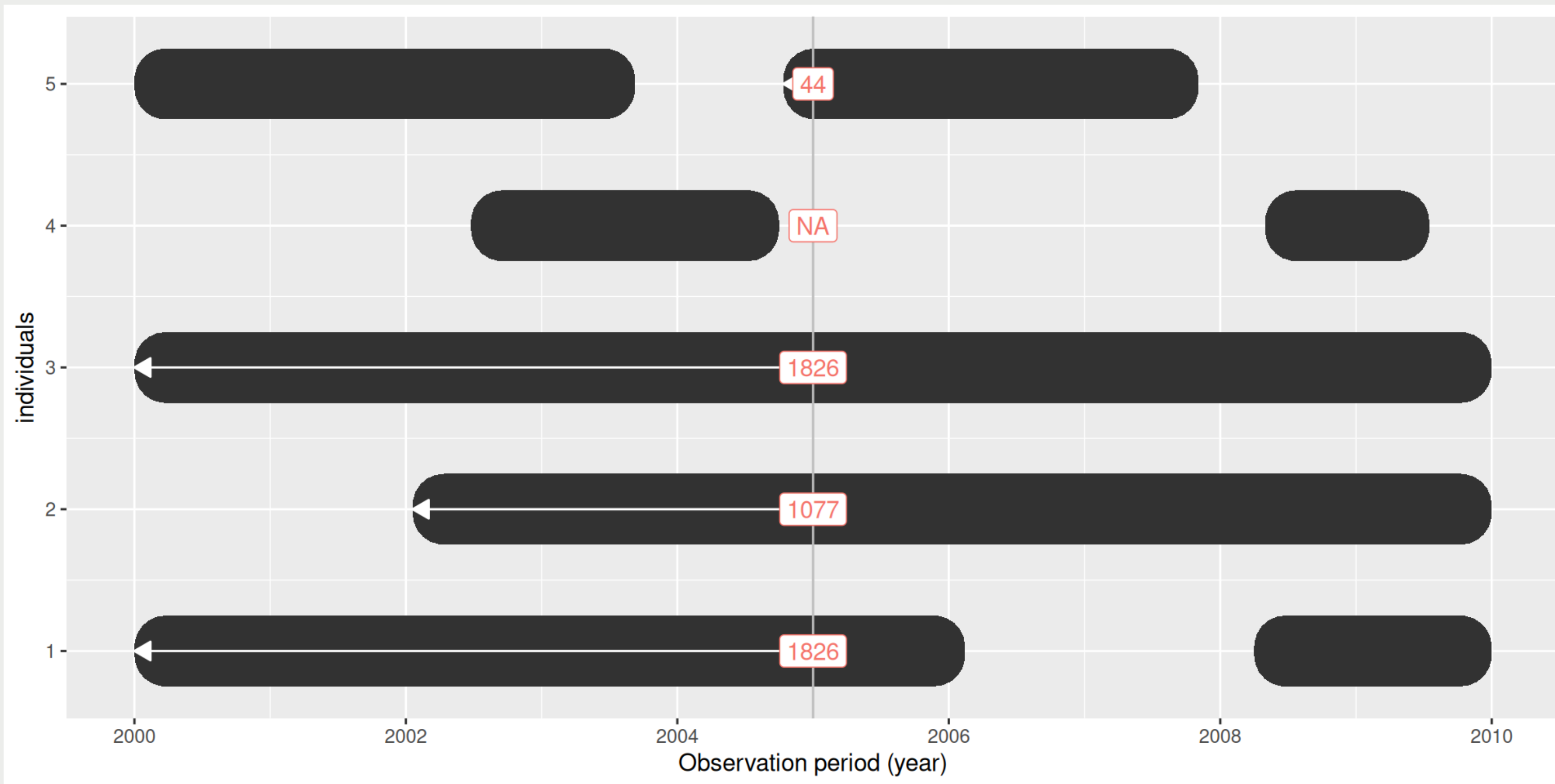
How much lookback time did each person have before the index date?



addPriorObservation()



addPriorObservation()



addPriorObservation()

The default output is the number of days from observation start to index date:

```
cdm$my_cohort |>
  addPriorObservation()
```

```
# A query:  ?? x 5
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
```

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	prior_observation
	<int>	<int>	<date>	<date>	<int>
1	1	84	2014-03-29	2014-03-29	14625
2	1	406	1996-06-27	1996-06-27	16199
3	1	1758	2002-09-28	2002-09-28	15380
4	1	1820	2014-08-17	2014-08-17	13340
5	1	2309	2016-10-10	2016-10-10	15900
6	1	4247	2018-03-30	2018-03-30	12795
7	1	4643	1944-06-22	1944-06-22	12850
8	1	4662	1994-06-10	1994-06-10	16015
9	1	648	1971-07-25	1971-07-25	14100

addPriorObservation()

You can return the observation-period start date instead of a duration:

```
cdm$condition_occurrence |>
  addPriorObservation(
    indexDate = "condition_start_date",
    priorObservationName = "start_observation", # name of the column
    priorObservationType = "date" # default = "days"
  ) |>
  select("person_id", "condition_occurrence_id", "condition_concept_id", "condition_start_date", "start_observation")
```

A query: ?? x 5

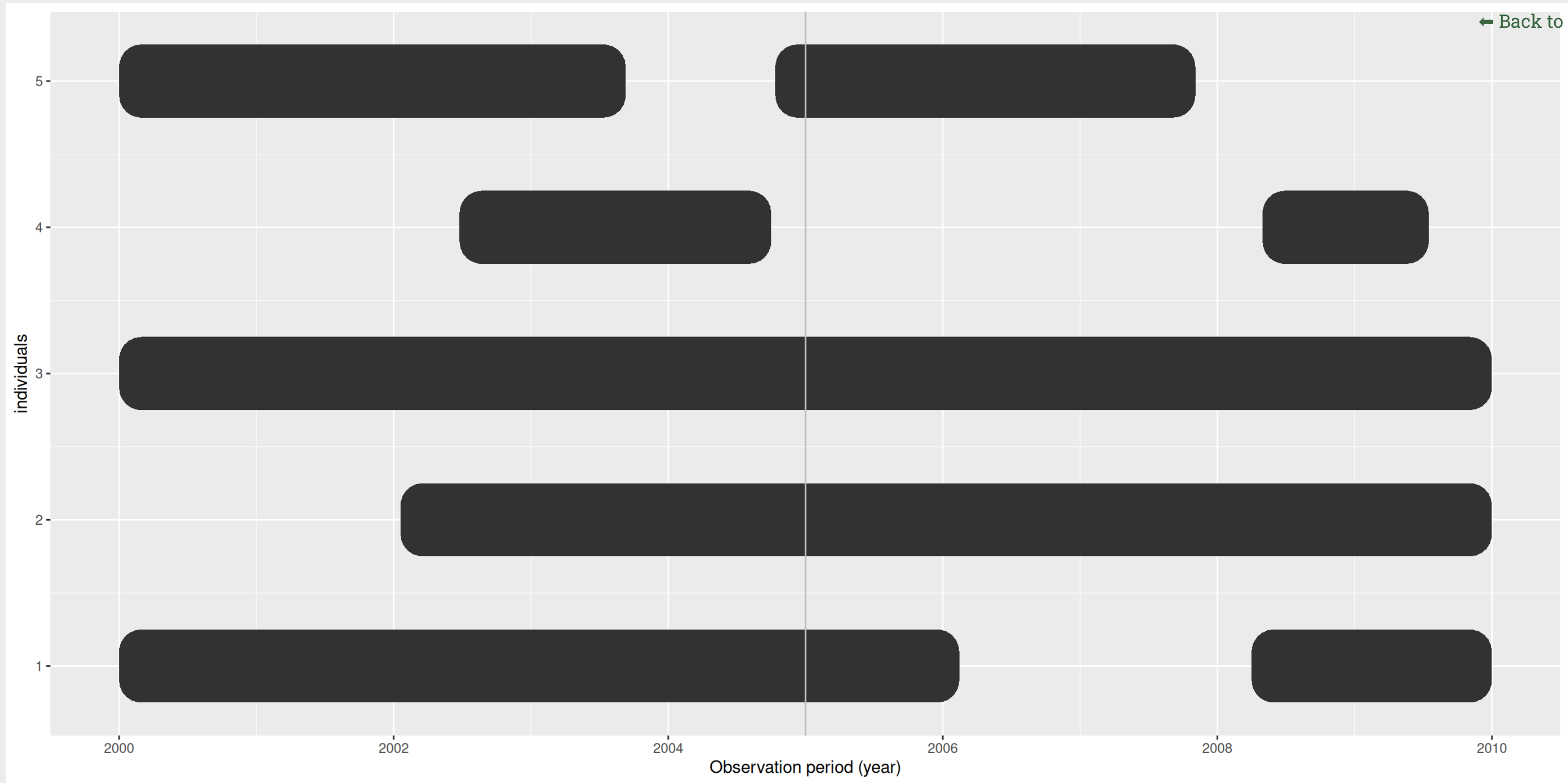
Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

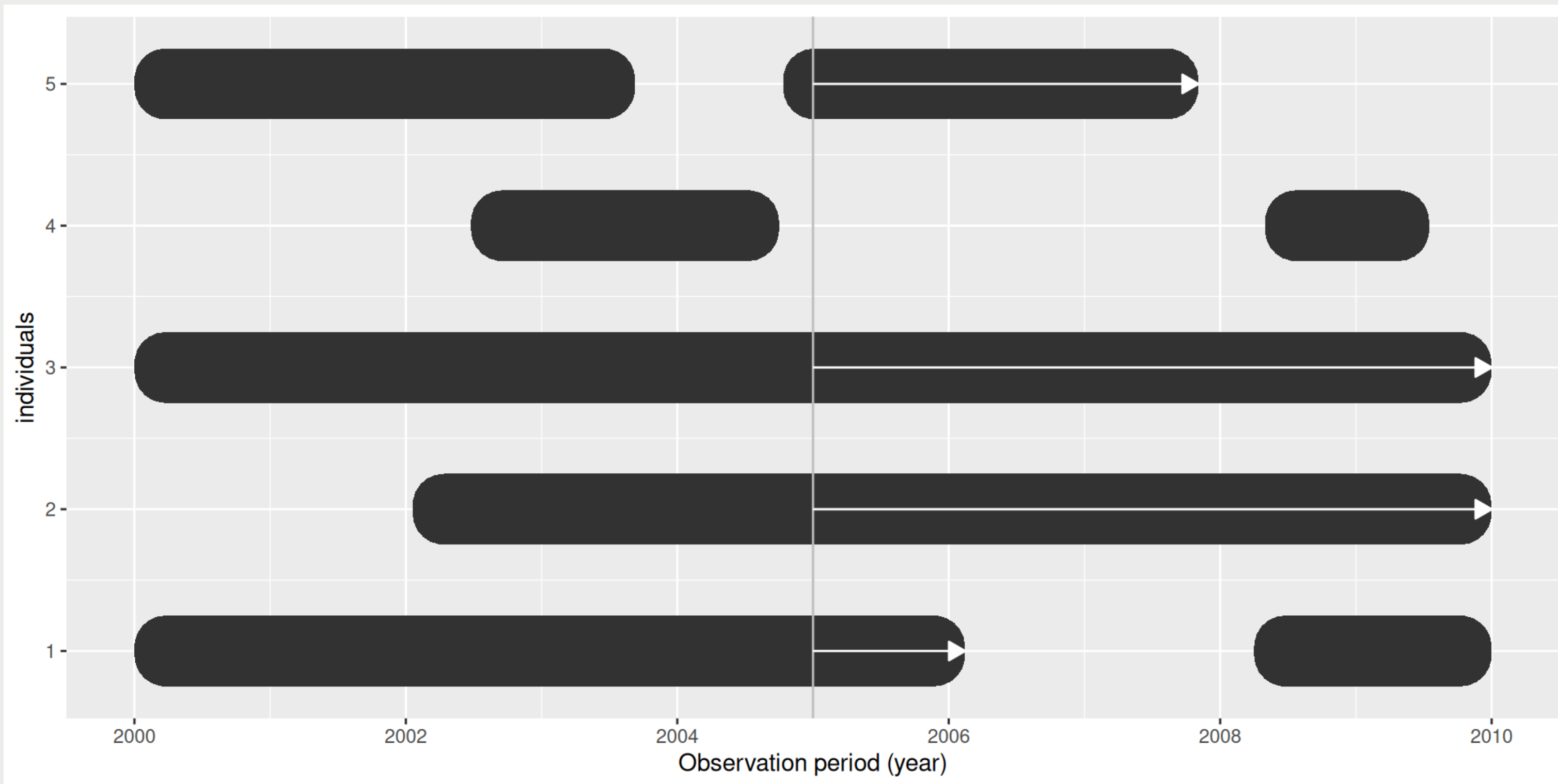
	person_id	condition_occurrence_id	condition_concept_id	condition_start_date	start_observation
	<int>	<int>	<int>	<date>	<date>
1	263	4483	4112343	2015-10-02	1954-06-20
2	273	4657	192671	2011-10-10	1975-09-14
3	283	4815	28060	1984-02-15	1980-09-01
4	293	4981	378001	2005-11-07	1964-10-13
5	304	5153	257012	1974-07-30	1956-05-07
6	312	5313	4134304	1991-05-14	1972-01-24
7	326	5513	28060	1979-09-23	1970-06-01
8	334	5655	40481087	1999-07-12	1935-06-06
9	341	5811	40481087	1990-09-14	1960-07-11

addFutureObservation()

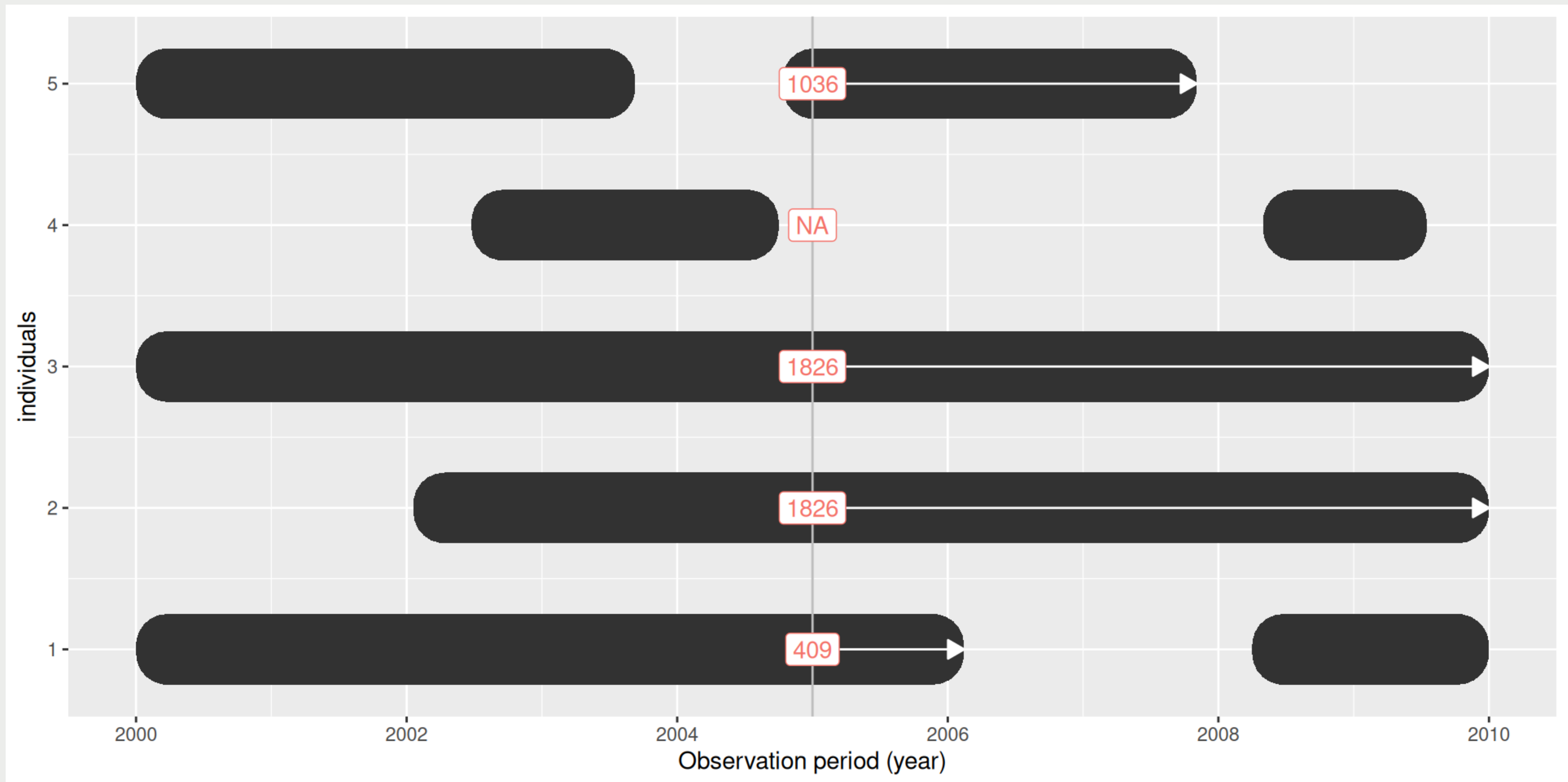
How much follow-up time did each person have after the index date? ...



addFutureObservation()



addFutureObservation()



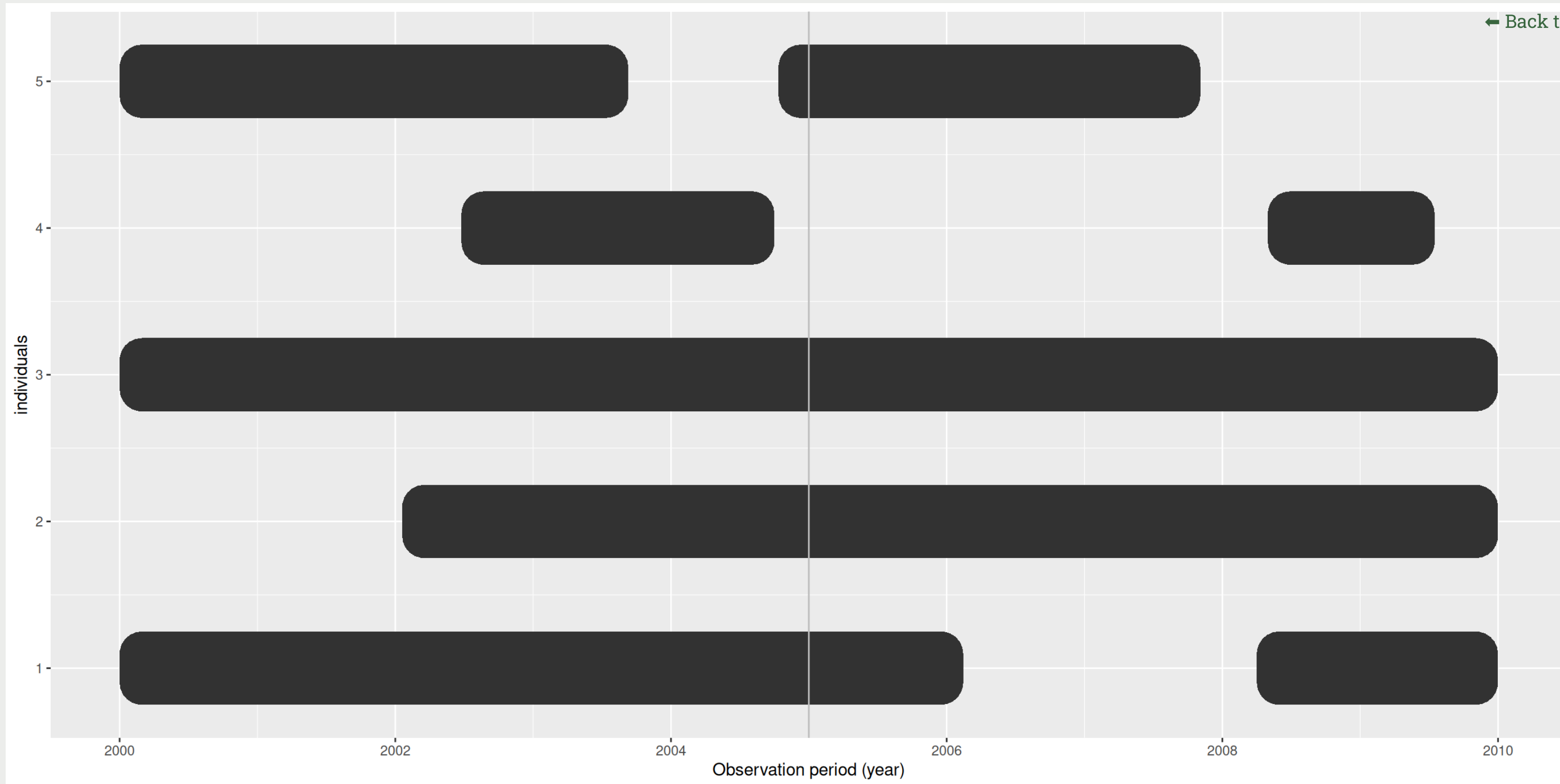
addFutureObservation()

```
cdm$my_cohort |>
  addFutureObservation()
```

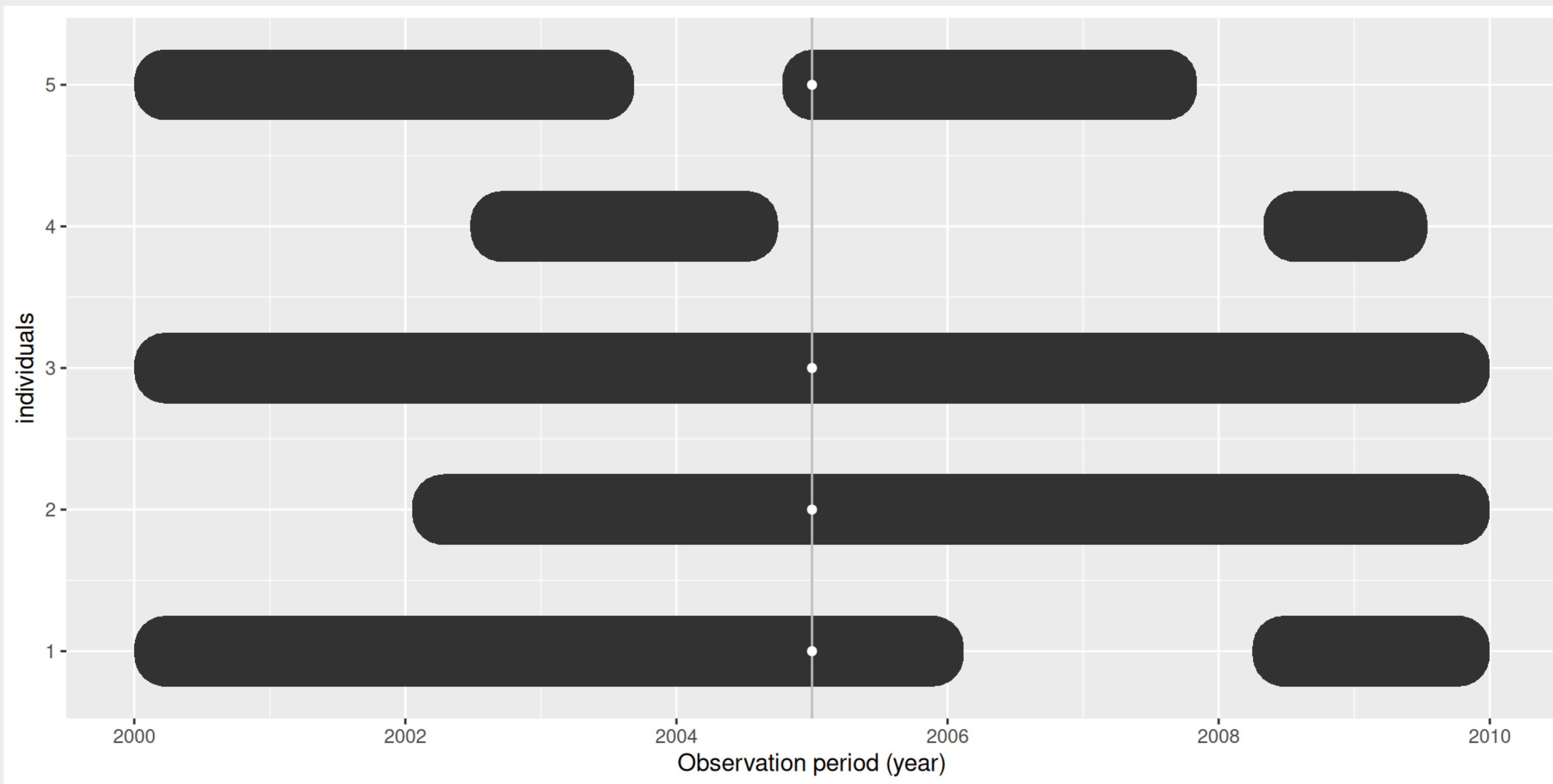
```
# A query:  ?? x 5
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
  cohort_definition_id subject_id cohort_start_date cohort_end_date future_observation
                <int>      <int> <date>                <date>                <int>
1                   1         84 2014-03-29            2014-03-29                1461
2                   1        406 1996-06-27            1996-06-27                8322
3                   1       1758 2002-09-28            2002-09-28                6005
4                   1       1820 2014-08-17            2014-08-17                1649
5                   1       2309 2016-10-10            2016-10-10                 928
6                   1       4247 2018-03-30            2018-03-30                 363
7                   1       4643 1944-06-22            1944-06-22               19644
8                   1       4662 1994-06-10            1994-06-10                8926
9                   1        648 1971-07-25            1971-07-25               17414
```


addInObservation()

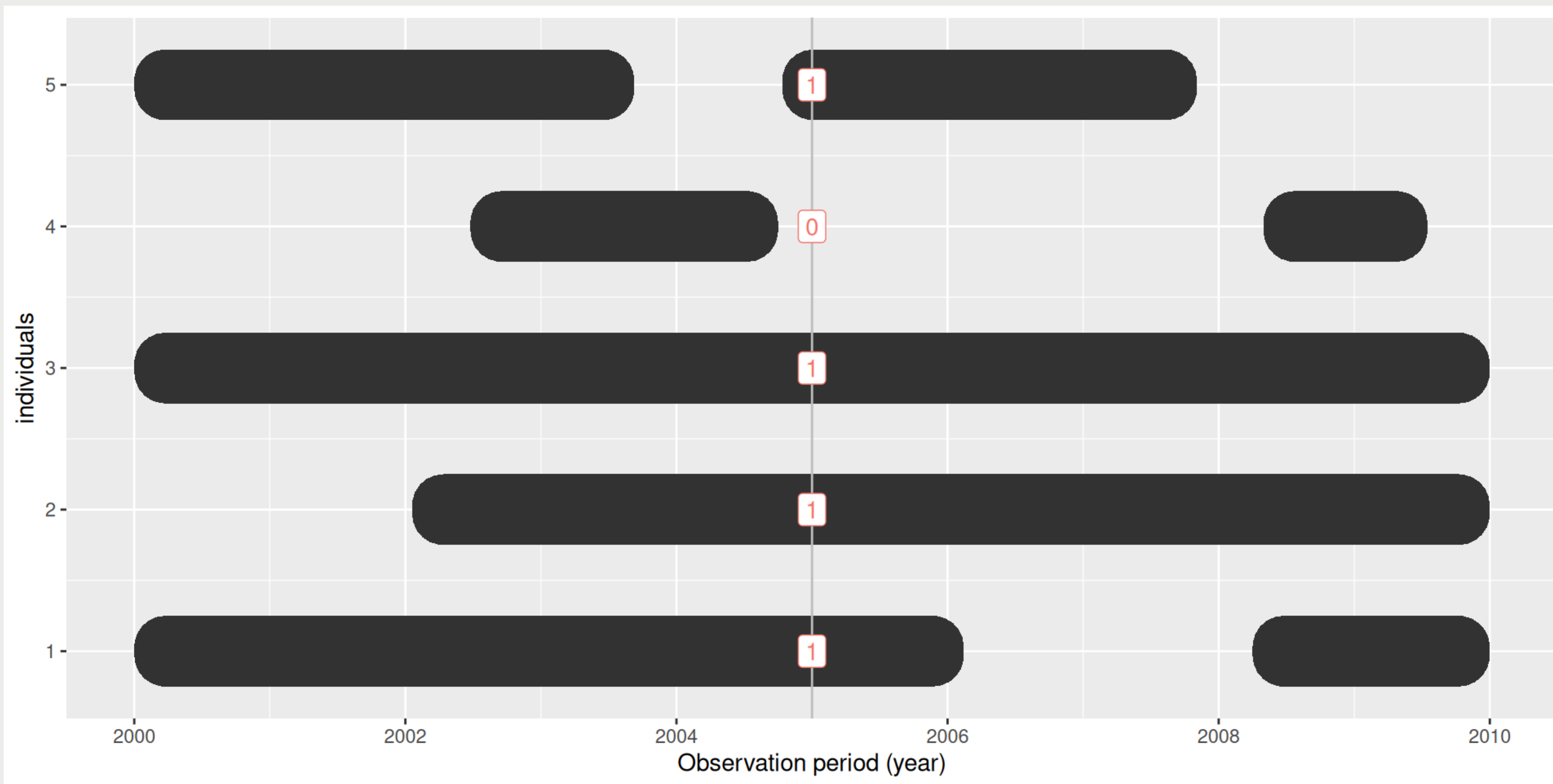
Is each index date inside an observation period?



addInObservation()



addInObservation()



addInObservation()

```
cdm$condition_occurrence |>
  addInObservation(indexDate = "condition_start_date") |>
  filter(in_observation == 1) |>
  select("condition_concept_id", "person_id", "condition_start_date", "in_observation")
```

```
# A query:  ?? x 4
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
  condition_concept_id person_id condition_start_date in_observation
              <int>      <int> <date>                <int>
1             4112343         263 2015-10-02              1
2             192671         273 2011-10-10              1
3              28060         283 1984-02-15              1
4             378001         293 2005-11-07              1
5             257012         304 1974-07-30              1
6             4134304         312 1991-05-14              1
7              28060         326 1979-09-23              1
8            40481087         334 1999-07-12              1
9            40481087         341 1990-09-14              1
--
```

addDateOfBirth()

Use `addDateOfBirth()` to add subjects' date of birth.

```
cdm$my_cohort |>
  addDateOfBirth()
```

```
# A query:  ?? x 5
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
  cohort_definition_id subject_id cohort_start_date cohort_end_date date_of_birth
            <int>         <int> <date>              <date>          <date>
1                   1           84 2014-03-29         2014-03-29      1974-03-14
2                   1          406 1996-06-27         1996-06-27      1952-02-19
3                   1         1758 2002-09-28         2002-09-28      1960-08-19
4                   1         1820 2014-08-17         2014-08-17      1978-02-07
5                   1         2309 2016-10-10         2016-10-10      1973-03-30
6                   1         4247 2018-03-30         2018-03-30      1983-03-19
7                   1         4643 1944-06-22         1944-06-22      1909-04-17
8                   1         4662 1994-06-10         1994-06-10      1950-08-05
9                   1          648 1971-07-25         1971-07-25      1932-12-16
--                   -
```


addBirthday()

Use `addBirthday()` to add specific birthdays of individuals, accounting for leap years and correcting the invalid dates to the next valid date

```
cdm$my_cohort |>
  addDateOfBirth() |>
  addBirthday(
    birthday = 5, # default is 0
    birthdayName = "5th_birthday"
  )
```

A query: ?? x 6

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	date_of_birth	`5th_birthday`
	<int>	<int>	<date>	<date>	<date>	<date>
1	1	84	2014-03-29	2014-03-29	1974-03-14	1979-03-14
2	1	406	1996-06-27	1996-06-27	1952-02-19	1957-02-19
3	1	1758	2002-09-28	2002-09-28	1960-08-19	1965-08-19
4	1	1820	2014-08-17	2014-08-17	1978-02-07	1983-02-07
5	1	2309	2016-10-10	2016-10-10	1973-03-30	1978-03-30
6	1	4247	2018-03-30	2018-03-30	1983-03-19	1988-03-19
7	1	4643	1944-06-22	1944-06-22	1909-04-17	1914-04-17
8	1	4662	1994-06-10	1994-06-10	1950-08-05	1955-08-05
9	1	648	1971-07-25	1971-07-25	1932-12-16	1937-12-16

addDemographics()

`addDemographics()` is a convenience wrapper for common patient-level features.

```
cdm$my_cohort |>
  addDemographics()
```

A query: ?? x 8

```
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
```

```
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
```

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	age	sex	prior_observation
	<int>	<int>	<date>	<date>	<int>	<chr>	<int>
1	1	6	2005-07-13	2005-07-13	41	Female	15170
2	1	123	1986-03-24	1986-03-24	35	Male	13130
3	1	129	2014-03-21	2014-03-21	39	Male	14410
4	1	16	2013-03-21	2013-03-21	41	Female	15134
5	1	65	2009-06-03	2009-06-03	42	Female	15405
6	1	74	2015-08-12	2015-08-12	43	Female	15925
7	1	42	1950-05-16	1950-05-16	40	Female	14804
8	1	187	1984-11-29	1984-11-29	39	Male	14374
9	1	18	2009-03-21	2009-03-21	43	Female	15830

addDemographics()

```
cdm$my_cohort |>
  addDemographics(
    age = TRUE,
    ageGroup = list("kids" = c(0, 17), "adults" = c(18, Inf)),
    sex = FALSE,
    priorObservation = TRUE,
    priorObservationName = "observation_start",
    priorObservationType = "date",
    futureObservation = TRUE,
    futureObservationName = "observation_end",
    futureObservationType = "date"
  )
```

A query: ?? x 8

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	age	age_group	observation_start
	<int>	<int>	<date>	<date>	<int>	<chr>	<date>
1	1	84	2014-03-29	2014-03-29	40	adults	1974-03-14
2	1	406	1996-06-27	1996-06-27	44	adults	1952-02-20
3	1	1758	2002-09-28	2002-09-28	42	adults	1960-08-19
4	1	1820	2014-08-17	2014-08-17	36	adults	1978-02-07
5	1	2309	2016-10-10	2016-10-10	43	adults	1973-03-30
6	1	4247	2018-03-30	2018-03-30	35	adults	1983-03-19
7	1	4643	1944-06-22	1944-06-22	35	adults	1909-04-17
8	1	4662	1994-06-10	1994-06-10	43	adults	1950-08-05
9	1	648	1971-07-25	1971-07-25	38	adults	1932-12-16

Your Turn

Starting from `cdm$my_cohort`, create a table with:

- sex
- age groups `<18`, `18-64`, and `65+`
- prior observation in days
- future observation in days

Your Turn

```
cdm$my_cohort |>
  addDemographics(
    age = FALSE,
    ageGroup = list(
      "<18" = c(0, 17),
      "18-64" = c(18, 64),
      "65+" = c(65, Inf)
    ),
    priorObservation = TRUE,
    priorObservationType = "days",
    futureObservation = TRUE,
    futureObservationType = "days"
  )
```

```
# A query: ?? x 8
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
```

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	age_group	sex	prior_observation
	<int>	<int>	<date>	<date>	<chr>	<chr>	<int>
1	1	6	2005-07-13	2005-07-13	18-64	Female	15170
2	1	123	1986-03-24	1986-03-24	18-64	Male	13130
3	1	129	2014-03-21	2014-03-21	18-64	Male	14410
4	1	16	2013-03-21	2013-03-21	18-64	Female	15134
5	1	65	2009-06-03	2009-06-03	18-64	Female	15405
6	1	74	2015-08-12	2015-08-12	18-64	Female	15925
7	1	42	1950-05-16	1950-05-16	18-64	Female	14804
8	1	187	1984-11-29	1984-11-29	18-64	Male	14374
9	1	18	2009-03-21	2009-03-21	18-64	Female	15830
...	...	...	...	...	...	...	...

Choosing a characteristic helper

Need	Helper
Sex	<code>addSex()</code>
Age or age groups	<code>addAge()</code>
Lookback time	<code>addPriorObservation()</code>
Follow-up time	<code>addFutureObservation()</code>
Date inside observation	<code>addInObservation()</code>
Common demographics together	<code>addDemographics()</code>

Add intersections

The intersection functions

- Did the person have an event?
- How many events did they have?
- When was the first or last event?
- How many days until or since the event?

15 functions

- addCohortIntersectFlag
- addCohortIntersectCount
- addCohortIntersectDays
- addCohortIntersectDate
- addCohortIntersectField
- addConceptIntersectFlag
- addConceptIntersectCount
- addConceptIntersectDays
- addConceptIntersectDate
- addConceptIntersectField
- addTableIntersectFlag
- addTableIntersectCount
- addTableIntersectDays
- addTableIntersectDate
- addTableIntersectField

How to choose

Source of target records	Flag
Cohort table	<code>addCohortIntersectFlag()</code>
Concept set	<code>addConceptIntersectFlag()</code>
OMOP table	<code>addTableIntersectFlag()</code>
Source of target records	Count
Cohort table	<code>addCohortIntersectCount()</code>
Concept set	<code>addConceptIntersectCount()</code>
OMOP table	<code>addTableIntersectCount()</code>

How to choose

Source of target records	Date
Cohort table	<code>addCohortIntersectDate()</code>
Concept set	<code>addConceptIntersectDate()</code>
OMOP table	<code>addTableIntersectDate()</code>
Source of target records	Days
Cohort table	<code>addCohortIntersectDays()</code>
Concept set	<code>addConceptIntersectDays()</code>
OMOP table	<code>addTableIntersectDays()</code>

How to choose

Source of target records	Field
Cohort table	<code>addCohortIntersectField()</code>
Concept set	<code>addConceptIntersectField()</code>
OMOP table	<code>addTableIntersectField()</code>

addCohortIntersectFlag

Did the person enter another cohort in the next 180 days or 181-365 days?

```
cdm$my_cohort |>
  addCohortIntersectFlag(
    targetCohortTable = "medications",
    window = list(c(1, 180), c(181, 365))
  ) |> glimpse()
```

Rows: ??

Columns: 6

```
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ subject_id          <int> 84, 406, 1758, 1820, 2309, 4247, 4643, 4662, 648, 1148, 1464, 2116, 2509, 3064,...
$ cohort_start_date   <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ cohort_end_date     <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ heparin_1_to_180    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
$ heparin_181_to_365  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0...
```

addTableIntersectCount

How many drug exposure records did the person have in the year after cohort entry?

```
cdm$my_cohort |>
  addTableIntersectCount(
    tableName = "drug_exposure",
    window = c(0, 365),
    nameStyle = "number_prescriptions",
    name = "my_cohort2" #Name of the new table, if NULL a temporary table is returned.
  )
```

```
# A query:  ?? x 5
# Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R
4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]
```

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	number_prescriptions
	<int>	<int>	<date>	<date>	<dbl>
1	1	84	2014-03-29	2014-03-29	1
2	1	406	1996-06-27	1996-06-27	1
3	1	1758	2002-09-28	2002-09-28	1
4	1	1820	2014-08-17	2014-08-17	2
5	1	2309	2016-10-10	2016-10-10	1
6	1	4247	2018-03-30	2018-03-30	1
7	1	4643	1944-06-22	1944-06-22	1
8	1	4662	1994-06-10	1994-06-10	1
9	1	648	1971-07-25	1971-07-25	1

addCohortIntersectDate

When did the person first enter the medication cohort after cohort entry?

```
cdm$my_cohort |>
  addCohortIntersectDate(
    targetCohortTable = "medications",
    targetCohortId = "heparin",
    window = c(0, Inf),
    order = "first", # date to use if there are multiple records for an individual during the window of interest
    nameStyle = "first_{cohort_name}"
  )
```

A query: ?? x 5

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	first_heparin
	<int>	<int>	<date>	<date>	<date>
1	1	84	2014-03-29	2014-03-29	NA
2	1	406	1996-06-27	1996-06-27	NA
3	1	1758	2002-09-28	2002-09-28	NA
4	1	1820	2014-08-17	2014-08-17	NA
5	1	2309	2016-10-10	2016-10-10	NA
6	1	4247	2018-03-30	2018-03-30	NA
7	1	4643	1944-06-22	1944-06-22	NA
8	1	4662	1994-06-10	1994-06-10	NA
9	1	648	1971-07-25	1971-07-25	NA
...	...	...	...	...	...

addConceptIntersectDays

How many days until the next record for a concept set?

```
cdm$my_cohort |>
  addConceptIntersectDays(
    conceptSet = list(tramadol = 1103314),
    window = c(0, 365),
    inObservation = FALSE, # If TRUE only records inside an observation period will be considered.
    nameStyle = "next_{concept_name}"
  ) |>
  glimpse()
```

Rows: ??

Columns: 5

```
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ subject_id           <int> 84, 406, 1758, 1820, 2309, 4247, 4643, 4662, 648, 1148, 1464, 2116, 2509, 3064,...
$ cohort_start_date    <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ cohort_end_date      <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ next_tramadol        <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA...
```


addConceptIntersectField

What is the last value of body weight recorded before index day?

```
cdm$my_cohort |>
  addConceptIntersectField(
    conceptSet = list(body_weight = 3012888),
    field = "value_as_number",
    window = c(-Inf, 0),
    order = "last",
    nameStyle = "body_weight_value"
  ) |>
  glimpse()
```

Rows: ??

Columns: 5

```
$ cohort_definition_id <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ subject_id          <int> 84, 406, 1758, 1820, 2309, 4247, 4643, 4662, 648, 1148, 1464, 2116, 2509, 3064,...
$ cohort_start_date   <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ cohort_end_date     <date> 2014-03-29, 1996-06-27, 2002-09-28, 2014-08-17, 2016-10-10, 2018-03-30, 1944-0...
$ body_weight_value   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,...
```

Your Turn

- Did the person have an heparin cohort entry in the year before index?
- How many condition occurrence records did they have in the year before index?

Your Turn

```
cdm$my_cohort |>
  addCohortIntersectFlag(
    targetCohortTable = "medications",
    targetCohortId = "heparin",
    window = list("prior_year" = c(-365, -1)),
    nameStyle = "prior_{cohort_name}"
  )

cdm$my_cohort |>
  addTableIntersectCount(
    tableName = "condition_occurrence",
    window = c(-365, -1),
    nameStyle = "prior_conditions"
  )
```

A query: ?? x 6

Database: DuckDB 1.5.4 [unknown@Linux 6.17.0-1018-azure:R

4.6.1//home/runner/work/RealWorldEvidenceSummerSchool2026/RealWorldEvidenceSummerSchool2026/Databases/GiBleed.duckdb]

	cohort_definition_id	subject_id	cohort_start_date	cohort_end_date	prior_heparin	prior_conditions
	<int>	<int>	<date>	<date>	<dbl>	<dbl>
1	1	84	2014-03-29	2014-03-29	0	2
2	1	406	1996-06-27	1996-06-27	0	1
3	1	648	1971-07-25	1971-07-25	0	1
4	1	1464	2002-08-13	2002-08-13	0	1
5	1	2116	1990-10-04	1990-10-04	0	1
6	1	2509	2006-05-20	2006-05-20	0	2
7	1	3064	1975-05-16	1975-05-16	0	2
8	1	4760	1995-03-06	1995-03-06	0	1
9	1	135	2009-11-27	2009-11-27	0	1

More intersection functions in PatientProfiles

- `addDeathDate`
- `addDeathFlag`
- `addDeathDays`

Resources

👉 [Package website](#)

👉 [CRAN](#)

👉 [PDF manual](#)

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